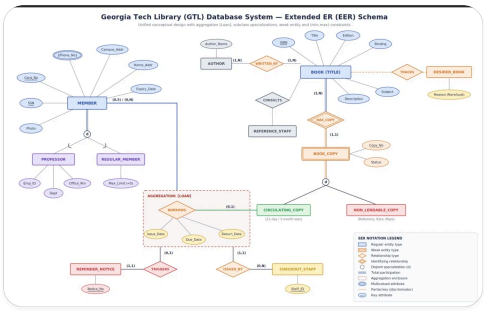
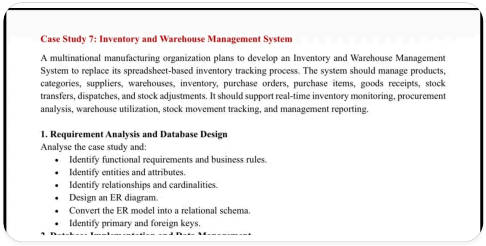
Simplified ER diagram and tables PDF

Share





Case Study 7: Inventory and Warehouse Management System

A multinational manufacturing organization plans to develop an Inventory and Warehouse Management System to replace its spreadsheet-based inventory tracking process. The system should manage products, categories, suppliers, warehouses, inventory, purchase orders, purchase items, goods receipts, stock transfers, dispatches, and stock adjustments. It should support real-time inventory monitoring, procurement analysis, warehouse utilization, stock movement tracking, and management reporting.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify functional requirements and business rules.
- Identify entities and attributes.
- Identify relationships and cardinalities.
- Design an ER diagram.
- Convert the ER model into a relational schema.
- Identify primary and foreign keys.

Simplified relations and and tables with proper ER diagram which matches the database Hey Claude, I have given the problem statement and the example of the ER diagram, how it should be looking like, and you have to create a PDF using this context, and the perfect problem statement is already given. So generate me a PDF that have this ER diagram and all the needed stuff which mentioned in the first line of this prompt. The first line is of the prompt is what the context given by the professor. So the PDF should contain everything. Before jumping straight into the development, just cross check the context and ask me some questions if you have any confusion.

Planning clarifying questions before starting development.

I've read both images. Here's how I understand the task:

- **Case Study 7: Inventory and Warehouse Management System**
- The main things to model are Product, Category, Supplier, Warehouse, Inventory, Purchase Order, Purchase Item, Goods Receipt, Stock Transfer, Dispatch, and Stock Adjustment.
- The PDF will include functional requirements, business rules, entities and attributes, relationships and cardinalities, the ER diagram, the relational schema, and the primary and foreign keys.
- Your professor wants the relations kept simple and the ER diagram to match the tables exactly.

Your screenshot ends just as **Part 2 (Database Implementation and Data Management)** begins, so I can't see what that part asks for. A few quick questions before I start:

hey, this is a assignment for a one person. So I want only the part one section of this case study. So build that PDF.

Read 2 files, ran a command ›

Got it, only Part 1. I'll use the same EER style as your example (weak entity, specialization, aggregation, and (min,max) labels) and keep the tables as structure only, no sample rows.

Ran 11 commands, read 6 files, and 3 more tools ›
Your PDF is ready. It is 11 pages and covers all of Part 1:

1. Problem overview
2. Functional requirements (14 of them)
3. Business rules (16 of them)
4. Entities and attributes
5. Relationships with (min,max) cardinalities
6. EER diagram, on a landscape page in the same style as your example
7. ER to relational mapping steps
8. 13 simplified relations, with a table for each one showing data types and constraints
9. Primary and foreign keys
10. A short check that each reporting need is covered

Before you submit:

- I put your name as Dinesh. The Roll No field is left blank for you to fill in.
- Every entity, relationship and attribute in the diagram has a matching table or column in the schema. That is what your professor asked for.

Design choices your professor may ask about in a viva:

- **Inventory is not a separate entity.** It is the M:N relationship between PRODUCT and WAREHOUSE, holding quantity on hand. It is shown as an aggregation, like the LOAN box in your example.
- **The four stock movement types are subclasses of STOCK_MOVEMENT.** These are goods receipt, dispatch, adjustment and transfer. This keeps the full stock history in one place, which covers the “stock movement tracking” requirement.
- **PURCHASE_ITEM is a weak entity.** It is owned by PURCHASE_ORDER and identified by PO_ID plus Line_No.
- **The order total is a derived attribute.** It is calculated from the items, so it is not stored in any table.

hey, I want to send this to someone of my friend, and he want to edit, like if he want to make changes, then what?

Figuring out the best way to share editable access.
A PDF is hard to edit. Your friend would need a paid tool like Adobe Acrobat, and even then, changing tables or the diagram is painful. It’s better to send him an editable version. There are two good ways:

- **Editable doc with a share link.** He opens the link in his browser, edits the text and tables, and can leave comments. Either of you can download it as Word or PDF when done.
- **Word file (.docx).** You send it on WhatsApp or email, and he edits it in Word or Google Docs.

The ER diagram is separate. In both options it would be a picture, so he can't move boxes or rename things in it. If he may need to change the diagram, I can also give you a draw.io file. He can edit it for free at diagrams.net and export a new image.

Which editable version do you want?

Doc with share link

Will your friend need to change the ER diagram too?

Yes, give me a draw.io file

Used Claude Docs integration, read a file ›

Outline is up. Now filling the sections.

Used Claude Docs integration ›

Now the functional requirements.

Used Claude Docs integration, ran 4 commands, and 4 more tools ›

The editable doc, Case Study 7: Inventory and Warehouse Management System (Part 1), is ready. It has everything from the PDF, and your friend can edit it, comment on it, or download it as Word or PDF.

How to share the doc: click **Share** at the top of the doc.

- If your friend is in your college's Claude workspace, add him by name and give him edit access.
- If he isn't, look in Share for a link option. If there's no link option, click the doc's name, choose **Export**, and send him the Word file.

How to change the diagram: send him `er_diagram.drawio`.

1. He opens it free at diagrams.net (File → Open).
2. The boxes are connected, so the lines and (min,max) labels move when he drags a box.
3. When he's done, he exports a PNG and replaces the image in Section 6 of the doc.

`er_diagram.png` is the same diagram as a plain image, in case he only needs the picture.

Download all

small doubt that free claude plan users also can edit and export right?

Checking permissions for sharing and plan access levels.
I don't know that for sure, so I'll check Anthropic's help pages.

Searched the web ›
Probably not. Anthropic's help page says Claude Docs is a beta feature on Pro, Max, Team and Enterprise plans, and it isn't available on the Free plan. Sharing also has limits. The person you share with needs their own Claude account. Anthropic's French help page adds that docs can't be shared by email invite outside your organization on any plan. Your college Team plan is an organization, so a friend outside it likely can't be added as an editor. support.claude.com +2

I couldn't find a page that says whether a free user can open or edit a doc someone else shared, so I wouldn't count on it.

The safe way is to send him a Word file, because anyone can open and edit one for free:

1. Open the doc and click its name at the top.
2. Choose **Export**, then **Word**.
3. Send him the file on WhatsApp or email, along with `er_diagram.drawio`.

He can edit the Word file in Word or Google Docs. The draw.io file opens free in a browser at diagrams.net, so he needs no Claude account for either.

Do you want me to make the Word file for you now?